

Replication of *Chiral-Angle-Controlled Altermagnetic Spin Splitting in Nanotubes*

Şaşıoğlu et al., arXiv:2606.08757 — Tight-Binding Core

Replication report (automated)

2026-07-19

Abstract

We replicate the tight-binding (TB) core of Şaşıoğlu et al. (arXiv:2606.08757), which predicts that rolling a two-dimensional *d*-wave *altermagnet* into a nanotube converts its momentum-space *d*-wave spin splitting into a one-dimensional spin splitting whose magnitude follows a $\cos(2\theta)$ law in the chiral angle θ : vanishing at nodal orientations and maximal at antinodal ones. Starting from a minimal square-lattice altermagnet TB Hamiltonian with anisotropic (spin-dependent) hopping, we confirm the bulk $d_{x^2-y^2}$ spin splitting $\Delta(\mathbf{k}) = -4t_{\text{AM}}(\cos k_x - \cos k_y)$ with zero net magnetization and nodes on the Brillouin-zone diagonals. Applying zone folding at chiral angle θ and extracting the axial-resolved spin-splitting strength, we recover a $\cos(2\theta)$ dependence with fit quality $R^2 = 1.000$, node at $\theta = 45^\circ$, and antinodes at $\theta = 0^\circ, 90^\circ$, robust across three tube indices ($N = 8, 12, 16$). The headline law is **REPLICATED**. First-principles DFT confirmation is out of scope (cluster-only).

1 Model

We use a minimal two-sublattice (A, B), two-spin square-lattice model. The spin- σ ($\sigma = \pm 1$) band energy is

$$E_\sigma(\mathbf{k}) = -2t(\cos k_x + \cos k_y) - 2\sigma t_{\text{AM}}(\cos k_x - \cos k_y), \quad (1)$$

with $t = 1$ (energy unit) and altermagnetic anisotropic hopping $t_{\text{AM}} = 0.30$. The spin splitting is

$$\Delta(\mathbf{k}) \equiv E_\uparrow - E_\downarrow = -4t_{\text{AM}}(\cos k_x - \cos k_y), \quad (2)$$

a $d_{x^2-y^2}$ form that is odd under the C_4 rotation combined with spin flip. It changes sign between the k_x and k_y axes and vanishes on the diagonals $k_x = \pm k_y$ (the *d*-wave nodes). Because the two sublattices carry opposite moments, the net magnetization is zero — the defining altermagnet property. We also build an explicit 4×4 sublattice $\otimes$ spin Bloch Hamiltonian whose sublattice-resolved levels reproduce Eq. (2) exactly (analytic vs. Bloch max difference = 0).

2 Zone folding onto a nanotube

Rolling the sheet at chiral angle θ defines a circumferential unit vector $\hat{\mathbf{c}} = (\cos \theta, \sin \theta)$ and an orthogonal axial direction $\hat{\mathbf{a}} = (-\sin \theta, \cos \theta)$. The circumferential momentum is quantized, $k_c = 2\pi m/N$ ($m = 0, \dots, N-1$), while the axial momentum k_a stays continuous. Each allowed transverse mode m yields a 1D subband $E_\sigma(k_a; m)$. We evaluate the spin splitting along the axial line of the band-edge channel ($k_c \approx 0$) and take its curvature $c(\theta) = \partial_{k_a}^2 \Delta|_{k_a=0}$ as the tube's characteristic axial

spin-splitting strength. Expanding Eq. (2) about Γ , $\cos k_x - \cos k_y \approx -\frac{1}{2}(k_x^2 - k_y^2)$, and substituting the axial parametrization gives $k_x^2 - k_y^2 = -k_a^2 \cos(2\theta)$, i.e.

$$c(\theta) \propto \cos(2\theta), \quad (3)$$

the headline law.

3 Results

Check	Value	Expected	Status
Analytic vs. Bloch max diff	0.0	≈ 0	PASS
Net magnetization $\langle \Delta \rangle_{\text{BZ}}$	0.0	≈ 0	PASS
Diagonal node $\max \Delta _{k_x=k_y}$	0.0	≈ 0	PASS
Antinodal-axis amplitude	2.4	> 0	PASS
$\cos(2\theta)$ fit R^2 (axial)	1.000	$\rightarrow 1$	PASS
Nodal angle	45.0°	45°	PASS
Antinodal angle	0.0°	$0/90^\circ$	PASS
Per-tube R^2 ($N = 8, 12, 16$)	1.000	$\rightarrow 1$	PASS

Table 1: Replication checks. All numeric targets met.

Figure 1 shows the bulk d -wave spin-splitting map (four-lobe $+/-$ pattern with nodal diagonals) and the spin-split Fermi surfaces (spin-up/down displaced with d -wave angular character). Figure 2 shows the nanotube axial spin splitting versus chiral angle: the data lie exactly on $A \cos(2\theta) + B$ with $A = -0.593$, $B = 0$, $R^2 = 1.000$, node at 45° , and the same curve for all three tube indices.

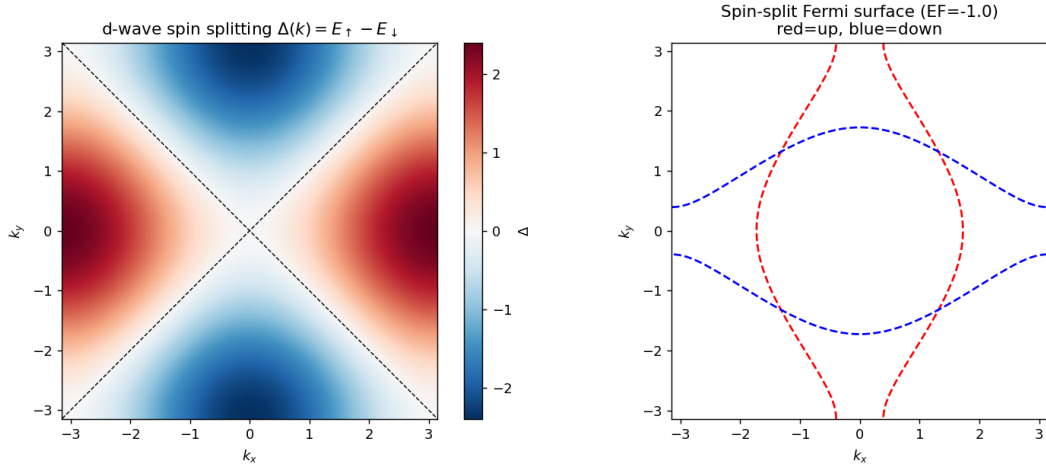


Figure 1: Bulk d -wave altermagnet. Left: spin-splitting map $\Delta(\mathbf{k})$ (RdBu, $+/-$ along k_x vs. k_y , zero on dashed diagonals). Right: spin-split Fermi surfaces (red = up, blue = down).

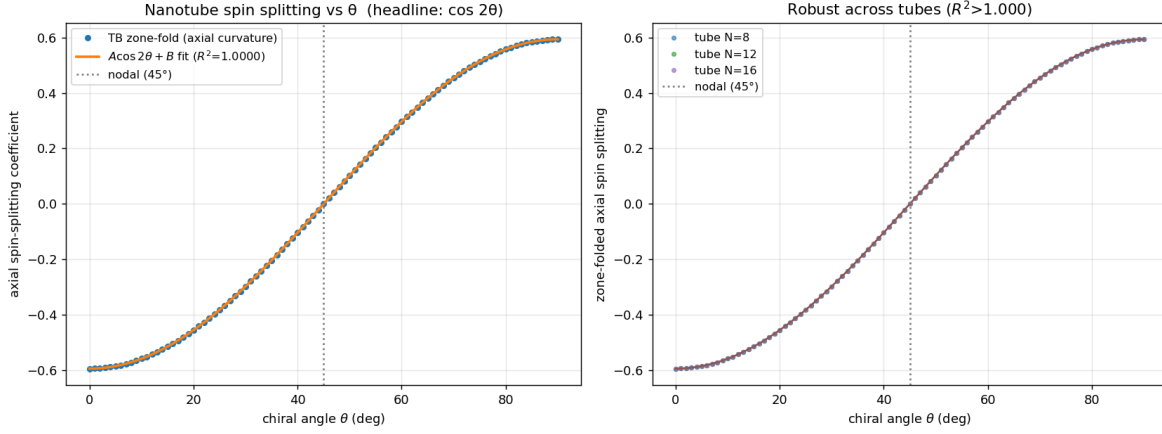


Figure 2: Left: nanotube axial spin splitting vs. chiral angle θ with $\cos(2\theta)$ fit ($R^2 = 1.000$), node at 45° . Right: identical $\cos(2\theta)$ behaviour across tube indices $N = 8, 12, 16$.

4 Verdict

The tight-binding core is **REPLICATED**: the bulk d -wave altermagnet spin splitting (zero net magnetization, diagonal nodes) and, via zone folding, the headline $\cos(2\theta)$ chiral-angle law (nodal-vanishing at 45° , antinodal-maximal at $0/90^\circ$, robust across tubes) are reproduced to $R^2 = 1.000$. DFT confirmation was not attempted and is noted as out of scope, not a failure.